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DETAIL SPECIFICATION

HEATING ELEMENTS, ELECTRICAL: CARTRIDGE, STRIP, AND TUBULAR TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cartridge, strip, and tubular type electrical heating elements used in various electrically-heated equipment (for example: immersion heaters for water or oil, cooking equipment, hospital equipment, and space heaters). Heating elements of these types are commonly used in industrial and domestic applications. Except where conflicts exist, industrial and commercial standards may be applied.

1.2 Classification. Heating elements are of the following types, as specified (see 6.2).

a. Type I – cartridge
b. Type II – strip
c. Type III – tubular

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-917 - Electric Power Equipment, Basic Requirements for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1399-300-1 - Low Voltage Electric Power, Alternating Current

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM B163 - Standard Specification for Seamless Nickel and Nickel Alloy Condenser and Heat-Exchanger Tubes

ASTM B344 - Standard Specification for Drawn or Rolled Nickel-Chromium and Nickel-Chromium-Iron Alloys for Electrical Heating Elements

ASTM B456 - Standard Specification for Electrodeposited Coatings of Copper Plus Nickel Plus Chromium and Nickel Plus Chromium

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Material not specified herein shall be selected on the basis of meeting the performance requirements of this specification at minimum cost.

3.1.1 Resistance wire. Resistance wire shall be nickel-chromium wire conforming to composition 60Ni-16Cr or 80Ni-20Cr of ASTM B344.

3.1.2 Insulating material. Granular insulating material shall be high purity magnesium oxide and shall be densely packed. Insulating material for formed parts, such as spacers, cores, or bushings shall be high quality electrical grade ceramic of high thermal conductivity, high insulation resistance, and formulated for physical strength.

3.1.3 Sheath materials. Sheath materials shall be corrosion-resisting in the surrounding medium and able to perform at the temperatures required by a particular application. Table I shall be used as a guide. Nickel-chromium plating of sheaths shall be in accordance with ASTM B456, classification Cu/Ni30d Cr r.

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TABLE I. Recommended sheath materials for various applications and temperatures.

Sheath material	Maximum sheath temperature °F (°C) 1/	Application 2/
Copper	350 (176.7)	A
Nickel-chromium-plated copper	350 (176.7)	A, B
Steel	750 (398.9)	B, C
Nickel-copper alloy	900 (482.2)	A, C, D, E
Nickel-silver	1,000 (537.8)	A, C, D, E
Chrome-steel	1,200 (648.9)	A, B, C, D, E
Corrosion-resisting steel	1,200 (648.9)	A, B, C, D, E
Nickel-iron-chromium alloy 3/ (Incoloy, or similar) UNS N08800	1,500 (815.6)	A, B, C, D, E
Nickel-chromium-iron alloy 3/ (Inconel, or similar) UNS N06600	1,600 (871.1)	A, B, C, D, E

NOTES:

1/ Operation above 1,000 degrees Fahrenheit (°F) (537.8 Celsius [°C]) for end sealed units is not permitted.

2/ Applications:

A – Water immersion

B – Oil immersion

C – Cast-in-metal

D – Contact metal heating

E – Air heating

3/ See ASTM B163.

3.1.4 Prohibited materials. The heating elements shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.1.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2 Type I, cartridge.

3.2.1 Description. Type I heating elements are metal-sheathed heating elements, round in cross-section, either closed at one end with terminals or lead wires at the opposite end, or terminals at both ends. The sheath encloses resistance wire which shall be carefully positioned in the insulating material. Bending or forming of cartridge type elements shall not be permitted.

3.2.2 Terminal construction. The terminal end of the heating element shall be constructed so as to minimize the entrance of atmospheric moisture into the heating element. The terminal construction shall satisfy performance requirements for the intended application, temperature, and atmospheric conditions present. Unless a hermetic, or other special type seal is specified (see 3.5 and 6.2), the terminal design and materials shall be as selected by the manufacturer to satisfy the requirements of a particular application. The type of connection, either threaded terminal studs or flexible leads of high temperature wire, shall be as specified (see 6.2). Nuts and lockwashers for making connections shall be furnished with heating elements having threaded terminal studs. Where provided, flexible leads shall be at least 10 inches long (25.4 centimeters).

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3.2.3 Insulation resistance, cold. Each heating element shall have a minimum insulation resistance of 25 megohms at room temperature (see 6.5).

3.2.4 Insulation resistance, hot. Each heating element shall have a minimum insulation resistance of 10 megohms at the maximum operating temperature for the sheath material (see 3.1.3 and 6.3.2) or actual operating temperature for the element (see 6.2). For heating elements whose maximum operating temperature is in excess of 1,000 °F (537.8 °C), the hot resistance may be tested with its entire system in accordance with MIL-DTL-917 as outlined in 4.2.6 of this specification.

3.3 Type II, strip.

3.3.1 Description. Type II heating elements are metal-sheathed heating elements, usually rectangular in cross-section with terminals projecting through the sheath, perpendicular to one major plane surface of the sheath. The sheath shall enclose helically coiled resistance wire uniformly spaced over the length and width of the element. Mounting slots or holes shall be provided at both ends of the heating element. Heating elements shall be capable of lengthwise bending by the manufacturer to a minimum inside radius of 8 inches (20.3 centimeters). Location and description of bends, if desired, shall be as specified (see 6.2). Strip type elements are not designed for use as immersion heaters and are not recommended for use in moist or humid locations or where terminals are located in an air stream.

3.3.2 Terminal construction. Terminals shall be of the threaded stud type and be designed to minimize the entrance of atmospheric moisture into the heating element. Nuts and lockwashers for making connections shall be furnished with the heating elements. The location of terminals for these heating elements are as follows:

a. One at each end.
b. Paralleled at one end except on ¾- and 1-inch widths.
c. In line at one end.
d. Offset or staggered at one end.
e. Three on one end for three heat.
f. Two in center.

Terminal location shall be as specified (see 6.2).

3.3.2.1 Moisture retardment. Each heating element shall have the terminals sealed with a silicone resin to prevent moisture from entering the insulating material during transit or storage. The compound shall be self-dissipating upon initial application of power.

3.3.3 Insulation resistance, cold. Each heating element shall have a minimum insulation resistance of 25 megohms at room temperature (see 6.5).

3.3.4 Insulation resistance, hot. Each heating element shall have a minimum insulation resistance of 10 megohms at the maximum operating temperature for the sheath material (see 3.1.3 and 6.3.2) or actual operating temperature for the element (see 6.2). For heating elements whose maximum operating temperature is in excess of 1,000 °F (537.8 °C), the hot resistance may be tested with its entire system in accordance with MIL-DTL-917 as outlined in 4.2.6 of this specification.

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3.4 Type III, tubular.

3.4.1 Description. Type III heating elements are metal-sheathed heating elements, formed from a section of tubing and have a terminal projecting at each end. The finished cross-section of the element need not be round. The sheath shall enclose a helically coiled resistance wire centered in magnesium oxide insulating material. High density of the insulating material shall be obtained by reducing the cross-section of the heating element under pressure. The heating element shall be bent or formed into the desired shape. Heating elements shall be capable of being bent by the manufacturer to a minimum inside radius equal to the sheath diameter. It is generally recommended that bends made by other than the heater manufacturer have an inside radius at least five times the sheath diameter. An unheated length shall be provided at the ends of the heating elements to prevent damage to the connections and wire by over-heating. Heating elements should not be bent within 1 inch (2.54 centimeters) of the unheated length unless approval to perform such bending is obtained from the manufacturer.

3.4.2 Terminal construction. The ends of the heating element and the terminals shall be constructed so as to minimize the entrance of atmospheric moisture. Terminal studs shall be of the threaded type unless brazed lugs are specified (see 6.2). Nuts and lockwashers for making connections shall be furnished with heating elements having threaded terminal studs. The end seal shall be of a type that satisfies the performance requirements for the intended application, temperature, and atmospheric conditions present. Unless a hermetic or other special type end seal is specified (see 3.5 and 6.2), the end seal design and materials shall be selected by the manufacturer to satisfy the requirements for the application.

3.4.3 Insulation resistance.

3.4.3.1 Insulation resistance computation. The insulation resistance of elements at any temperature from room ambient temperature to 1,200 °F (648.9 °C) shall be not less than the value computed by the following:

$$R = (50 - 0.02T)/L$$

Where R = Insulation resistance in megohms

T = Temperature of element sheath in °F or °C

When measuring insulation resistance (cold) (see 4.2.5), T shall be the room temperature

L = Length of element (end to end, feet)

3.4.3.2 Insulation resistance measurements. The insulation resistance measurements required by 4.2.5 shall be made at room temperature (see 6.5) and at maximum operating temperature for the sheath material (see 3.1.3) or actual operating temperature (see 6.2), whichever is the least. For heating elements whose maximum operating temperature is in excess of 1,000 °F (537.8 °C), the hot resistance may be tested with the entire system in accordance with MIL-DTL-917 as outlined in 4.2.6 of this specification.

3.5 End seals.

3.5.1 Hermetic end seals for type I and type III elements. Hermetic end seals are superior to other end seals in resisting atmospheric moisture, fumes, grease, and water. Hermetic end seals also have greater electrical creepage than most other types. For type I and type III elements in applications where these factors are important considerations, it is recommended that hermetic seals be specified for applications where temperatures do not exceed 1,000 °F (537.8 °C) (see 6.2). For example, on duct heaters used in air conditioning systems, moisture is apt to condense on the heating elements during periods when heat is not required, and cool air is circulating in the duct. In this case, a hermetic end seal would be advised.

3.5.1.1 Requirements for hermetic seals. The creepage path along the insulated portion of any hermetic seal shall be a minimum of 5/16 inch (8 millimeters) long for a voltage of 300 Volts or less. For more than 300 Volts, the length of the creepage path shall be a minimum of ½ inch (13 millimeters) long.

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3.5.1.2 Ceramic-to-metal hermetic seals for type III elements. On type III elements where the temperature of the heating element end exceeds 250 °F (121.1 °C), hermetic seal shall be of the ceramic-to-metal type, using a high-grade glazed alumina ceramic as the insulator. Ceramic-to-metal type seals shall be brazed to the heating element sheath.

3.5.1.3 Moisture resistance of hermetically sealed terminals. Hermetically sealed terminals shall meet the requirements of 3.2.3 or 3.4.3, as applicable, for insulation resistance and the requirement of 3.9 for dielectric strength after the element has been completely immersed in water for a period of 10 minutes.

3.5.2 End seals for commissary cooking equipment. Heating elements for commissary cooking equipment shall be constructed so as to permit any trapped moisture to escape. Elements shall be internally dried before presenting for shipment. Maximum operating temperature shall be specified by the manufacturer when limited by the end seal.

3.5.3 Moisture retardment. Unsealed heating elements shall have the terminals sealed with a silicone resin to prevent moisture from entering the insulating material during transit or storage. The resin shall be self-dissipating upon initial application of power.

3.6 Special features. When specified (see 6.2), heating elements shall be provided with special features, examples of which are:

a. Mounting bushings, screw plug, or mounting flanges to facilitate sealing cartridge or tubular heaters in tank walls and so forth.

b. Fins (for forced air heaters).

Parts such as bushing, flanges, or fins shall be securely brazed to the heating element sheath.

3.7 Wattage. The wattage of the heating element at rated voltage shall be within the range of plus 5 to minus 10 percent of that specified (see 6.2).

3.8 Voltage. The operating voltage shall be either 440 or 115 Volts in accordance with MIL-STD-1399-300-1, as specified (see 6.2). The element shall operate satisfactorily on voltages up to 105 percent of the specified rating. When heating elements are intended for operation on a voltage other than line voltage (by series or wye connections), the line-to-line voltage shall also be specified. The heating element shall be insulated for the line voltage and shall have electrical creepage and clearance dimensions for line voltage.

3.9 Dielectric strength. Heating elements shall withstand a dielectric test of twice rated line voltage plus 1,000 for a period of 1 minute. The voltage wave shall approximate a sine wave and the frequency shall be not less than 60 hertz.

3.10 Descriptive information. The following information shall be as specified (see 6.2).

a. Sheath material.

b. Sheath diameter or cross-section dimensions.

c. Length of heated portion of element.

d. Length of sheath.

e. Overall length of element.

f. Location and description of bends, if required for strip or tubular elements.

g. Details of special features such as length of finned portion, fin diameter, distance between flanges of mounting bushings, sizes of mounting bushings, flanges, and so forth.

h. Dimensional and bending tolerances, if critical. Tolerances, if specified shall be as relaxed as application permits. The manufacturer’s normal manufacturing tolerances shall apply if no limits are specified.

i. Application data, if it is desired that the heater manufacturer select the heating element to best suit the particular application. This shall include data on material to be heated, rate of heating, air velocities, dimensional limitations, and so forth.

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4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall include the examination of 4.2.1 and the tests of 4.2.2 through 4.2.6.

4.1.1 Inspection lot. All heating elements of the same power rating and type, made for the same application (see 3.1.3), and offered for delivery at one time shall be considered a lot for purposes of sampling and inspection.

4.1.2 Sampling. Sample heating elements shall be selected from each lot in accordance with inspection level II of ANSI/ASQ Z1.4 and each of these elements shall be subjected to each of the examinations and tests specified in 4.2.1, 4.2.2, 4.2.3, 4.2.4, and 4.2.5, and the results of each test shall be compared with the requirements of this specification. Failure to conform to this specification for any of the examination or tests shall because for rejection. The acceptable quality level (AQL) for lots shall be as specified (see 6.2).

4.1.3 Sampling for insulation resistance (hot) tests. Sample elements shall be selected from each lot in accordance with inspection level S-4 of ANSI/ASQ Z1.4 and shall be subjected to the test of 4.2.6. If any element fails this test, it shall be cause for rejection. The acceptable quality level (AQL) for lots shall be as specified (see 6.2).

4.2 Test procedures.

4.2.1 Dimensions, materials, and workmanship. Each of the heating elements shall be subjected to a thorough visual and dimensional examination to ascertain that the materials, workmanship, design, and dimensions conform to this specification.

4.2.2 Wattage. The wattage shall be measured to determine compliance with 3.7.

4.2.3 Moisture resistance (elements having hermetically sealed terminals only). Where hermetically sealed terminals are required, each element shall be immersed in water for 10 minutes and the terminals wiped dry prior to the dielectric test of 4.2.4 and the insulation resistance test of 4.2.5.

4.2.4 Dielectric strength. The heating element shall be subjected to a dielectric test voltage of twice the rated line voltage plus 1,000 Volts. It shall be applied between the terminal and the sheath as specified in 3.9. The test shall be conducted with the heating element at room temperature (see 6.5). The heating element shall withstand the test for a period of 1 minute.

4.2.5 Insulation resistance, cold. The insulation resistance shall be measured with 500 Volts Direct Current (VDC) at room temperature (see 6.5) to determine compliance with 3.2.3, 3.3.3, or 3.4.3.

4.2.6 Insulation resistance, hot. The insulation resistance shall be measured with a minimum of 500 VDC within 5 seconds after operation of the element at maximum operating temperature for the sheath material (see 3.1.3), or end seal material (see 3.5), or operating temperature of the element (see 6.2), whichever is the least, to determine conformance with 3.2.4, 3.3.4, or 3.4.3. To prevent the element from being cooled before the test, the ambient fluid shall not be circulated after the power is removed. Heating elements with operating temperatures above 1,000 °F (537.8 °C) may be tested in the heater.

4.3 Prohibited materials. Prohibited materials shall be verified as specified (see 3.1.4 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but it not mandatory.)

6.1 Intended use. This specification covers cartridge, strip, and tubular types of electrical heating elements intended to be used as listed below aboard United States naval surface ships and submarines and ensures compatibility and proper integration relative to size, function, performance, and standardization.

a. Type I, cartridge:
   (1) Immersion heating
   (2) Contact metal heating

b. Type II, strip:
   (1) Contact metal heating

c. Type III, tubular:
   (1) Immersion heating
   (2) Cast-in-metal
   (3) Contact metal heating
   (4) Air heating

Electrical heating elements procured to this specification are form, fit, and function replacements for prior revisions, therefore, existing stock of electrical heating elements manufactured to previous version of document dated 9 September 1982 is acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of heating element required (see 1.2).

c. Requirement for prohibited materials verification (see 3.1.4 and 4.3).

d. Whether hermetic or other special type end seals are required for type I or type III heating elements (see 3.2.2, 3.4.2, and 3.5.1).

e. Whether type of connection for type I heating elements is to be threaded terminal studs or flexible leads (see 3.2.2).

f. Actual operating temperature if required (see 3.2.4, 3.3.4, 3.4.3.2, and 4.2.6).

g. Location and description of bends if desired, for type II and type III heating elements (see 3.3.1 and 3.10).

h. Location of terminals for type II heating elements (see 3.3.2).

i. Whether brazed lug type terminals are required in lieu of standard threaded studs for type III heating elements (see 3.4.2).

j. Special features (see 3.6).

k. Wattage (see 3.7).

l. Voltage (see 3.8).

m. Descriptive information, details of special features, or application data (see 3.10).

n. Dimensional tolerances, if required (see 3.10).

o. AQL requirements (see 4.1.2 and 4.1.3). Recommend an AQL of 1.5 percent for examination and tests specified in 4.2.1, 4.2.2, 4.2.3, 4.2.4, and 4.2.5. Recommend an AQL of 4.0 percent for test specified in 4.2.6.

p. Packaging requirements (see 5.1).

q. Responsible activity for performing inspection requirements, if other than the contractor. State whether the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (see 6.4).

r. Requirements for conformance inspection test procedures and/or test reports, if required (see 6.6).

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6.3 User information.

6.3.1 Vibration and shock. Use of fragile materials and design features that would cause the elements to be unusually susceptible to physical damage is to be avoided. The heating elements by themselves need not meet any specific vibration or shock requirements. Specific shock and vibration requirements may be specified for the equipment in which the elements are ultimately installed, in which case, the installed elements are to operate satisfactorily under the specified shock and vibration.

6.3.2 Heating elements used in applications above 1,000 °F (537.8 °C). Heating elements used above 1,000 °F (537.8 °C) which were tested in the heater (see 4.2.6) should be either insulated from the ship's hull or should be isolated from the distribution system.

6.4 Responsibility for inspection and compliance. Requirements for inspection, and compliance are as specified (see 6.2).

6.5 Definition.

a. Room temperature is defined as 68 to 77 °F (20 to 25 °C).

6.6 Conformance inspection reporting. Requirements for providing test procedures and test reports are as specified (see 6.2).

6.7 Subject term (key word) listing.

a. Heater

b. Resistance-wire

6.8 Changes from previous issue. Marginal notes are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH

Preparing activity:
Navy – SH
(Project 4540-2020-001)

Review activities:
Army – MD, MI
Navy – MC, MS, YD
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